STAT/MA 41600

In-Class Problem Set #3: August 31, 2026
Solutions by Mark Daniel Ward

Problem Set 3 Answers

1a. The probability that none of them prefer Claude is $(1 - 0.17)^5 = (0.83)^5 = 0.3939$.

1b. For all five of them to agree on the same preferred assistant, they must all prefer ChatGPT, or all prefer Gemini, or all prefer Claude. These are 3 distinct events, so the probability is $(0.50)^5 + (0.33)^5 + (0.17)^5 = 0.0353$.

2a. On Problem Set 2, we learned that the probability of a good result is 23/36. So the probability that the first 3 rolls of the pair of dice are all good results is $(23/36)^3$.

2b. The probability of a bad result, i.e., getting a sum of eleven on a roll, is 2/36. The probability of a bad result before a good result is $\frac{2/36}{2/36+23/36} = 2/25$.

3a. The probability of no green bears is $(5/6)^3 = 125/216$. So the probability of at least one green bear is $1 - (5/6)^3 = 91/216$.

3b. We note that $A \cup B$ is the event that there is at least one green bear or at least one purple bear. The event $(A \cup B)^c$ is the event that there are no green bears and no purple bears, so $P((A \cup B)^c) = (4/6)^3 = 8/27$. It follows that $P(A \cup B) = 1 - 8/27 = 19/27$.

4a. The probability that there are exactly five rainy days is $\binom{5}{5}(1/3)^5 = 1/243$. The probability that there are exactly four rainy days is $\binom{5}{4}(1/3)^4(2/3) = 10/243$. The events are disjoint so we can add them. So the probability that Bob feels sad about the weekly weather forecast is $1/243 + 10/243 = 11/243$.

4b. The probability that there are exactly three rainy days is $\binom{5}{3}(1/3)^3(2/3)^2 = 40/243$. So the probability that Mary feels sad about the weekly weather forecast is $1/243 + 10/243 + 40/243 = 51/243$.

5. Regardless of where Bob sits, the probability that Catherine is next to him is 2/3, so $P(V) = 2/3$. Similarly, we have $P(W) = 2/3$.

In fact, either V and W both occur (and, thus, $V \cap W$ occurs), or neither occurs. So it follows that $P(V \cap W) = 2/3$.

We conclude that $P(V \cap W) \neq P(V)P(W)$, so V and W are not independent. In fact, V happens if and only if W happens in this problem!